RISK MODIFICATION: TRANSFERRING, SHIFTING, AND HOW TO CHOOSE

Learning Outcome

- describe methods for measuring and modifying risk exposures and factors to consider in choosing among the methods

Risk transfer is the process of passing on a risk to another party, often, but not always, in the form of an insurance policy. Insurance is a legal contract in which one party, the insurer, agrees to cover some percentage of a loss incurred by another party, the insured, from a specific event in return for the payment of a premium by the insured. Insurance as a method of risk modification has been in existence for very long time, and in fact, is even mentioned in the Code of Hammurabi almost 4,000 years ago. Insurance has been widely used in the commercial shipping and farming industries going back hundreds of years. Insurance is almost as old as commerce itself.[25]

From the point of view of the insurer, insurance almost always works on the basis of diversification or pooling of risks. An insurer attempts to sell many policies with risks that have low correlations. The insurer assesses the pooled risks and charges a premium that covers the expected aggregate losses and the insurer's operating costs as well as leaves a profit. Insurers need accurate statistics on aggregate risks, but these are often not difficult to obtain. These actuarial data are widely available on accidents, illnesses, property and casualty damage, and death. In principle, a well-diversified insurer does not care if a single insured party has significantly larger-than-average claims as long as there is no reason to believe that the claims are correlated. There will be other parties that have smaller-than-average claims.

Insurers do have to manage their risks carefully. Some risks can be correlated. In the US Gulf Coast region, property insurance, which includes coverage for loss by hurricanes, is typically more expensive than property insurance in other regions. Even with a higher premium, an insurer has to avoid providing too much property coverage in an area where a systemic event, such as a hurricane, can occur in order to diversify its risk exposure.

Although insurers carefully assess their risk and charge premiums that they believe accurately reflect expected losses, they nonetheless remain responsible for potentially large claims. Insurers also manage their risk by avoiding writing too many policies with similar and potentially correlated risks and by selling some of the risk to another insurer, a practice known as reinsurance. A company that primarily insures property in the US Midwest, which is highly subject to tornado risk, might be willing to accept some Gulf Coast hurricane risk for a reasonable premium. Insurers often write provisions into contracts to exclude coverage of special cases. For example, a war might nullify insurance coverage in an area. Most insurance policies also contain provisions to guard against moral hazards, such as suicide or destroying one's own property. In the last 20 years or so, some insurance companies have issued bonds that permit them to legally avoid paying principal and/or interest if insurance claims exceed a certain amount. These instruments, known as catastrophe bonds, essentially pass some of the insurance risk on to the investors who buy the bonds.

Most insurance policies do not cover *all* of the risk that is insured. It is common for policies to contain a provision known as a deductible. A deductible is a monetary amount of the loss that will be covered by the insured before any claims are paid. Thus, both the insured and the insurer bear some of the risk, although the insurer usually bears the greater amount. Deductibles serve several purposes. Because insurers incur fixed costs for each claim, deductibles reduce the number of small claims. Deductibles also encourage good

As noted, the concept of insurance relies on the diversification or pooling of risks. In a few cases, however, the risks are not easy to pool. For example, suppose a volatile but extremely successful actor is signed to star in a movie. The production company knows that it runs the risk that the actor will engage in behavior that damages the ability of the company to finish the movie. The number of volatile and extremely successful actors for whom policies could be written at the same time is somewhat limited. Thus, an insurer would have to bear that risk without the benefit of diversification.

For example, suppose a television network plans to cover the Olympics but is concerned about a possible cancellation or boycott. It might want an insurance policy to cover it against loss. Specialized coverage is possible through such companies as Lloyd's of London. The approximately 350-year old Lloyd's is famous for covering unusual risks. It does so by organizing groups of investors who are willing to bear a risk for a premium. These groups, called syndicates, are subject to the full extent of losses. In many cases, investors in these syndicates have been required to pay substantial amounts of money to cover losses.[26] These examples illustrate how syndicates work. Although there is only one Olympics to insure, there may also be only one actor to insure. Because the two risks are uncorrelated, a company could write policies on both risks and would achieve some diversification. Moreover, there are other unusual risks that can be covered such that the aggregate pool would represent a highly diverse set of risks that have low correlations.

A very slight variation of insurance is a surety bond. With a surety bond, an insurer promises to pay an insured a certain amount of money if a third party fails to fulfill its obligation. For example, if a party engages the services of another party, the first party is covered if the party obligated to provide the service fails to perform to a satisfactory degree. Surety bonds are widely used in commercial activity when one party bears the risk of the potentially high cost of non-performance by another party. A slight variation of a surety bond is a fidelity bond, which is often used to cover against losses that result from employee dishonesty. Bonds of this type work very similarly to insurance and rely on the pooling of uncorrelated risks.[27] Other similar arrangements include indemnity clauses and hold harmless arrangements, such as when two parties sign a contract and one party agrees to hold the other harmless and/or indemnify the other in the event of loss.

The use of insurance by so many as a risk management tool suggests that the cost of risk exceeds the actuarial cost to many individuals and enterprises. *Ex ante* consideration of the cost of a risk in terms of the organization's value or utility ties risk mitigation back to the risk tolerance decision and the most fundamental governance decisions on which value-added strategies to pursue. As an alternative to *ignoring* the cost of risk, the impact on enterprise value should be quite positive.

Risk Shifting

Whereas risk transfer refers to actions taken that pass the risk on to other parties, **risk shifting** refers to actions that change the distribution of risk outcomes. Risk transfer is often associated with insurance, whereas risk shifting generally involves derivatives as the risk modification vehicle. Although insurance is a form of risk management based on the pooling or diversification of risks, risk shifting diverts some portion of the risk distribution to another market participant who either bears the risk or intermediates that risk by moving it to yet another party. The organization may want to adjust its probability distribution of returns, essentially adjusting the payoff diagram of its risk exposures. An example is a company that is willing to make slightly less profit than it otherwise would if the stock market is up to prevent it from losing too much money if the stock market is down, for example, more than 20% next year. It is adjusting its potential economic outcomes by shifting the probability distribution of its profits conditional on market performance. Risk shifting represents the bulk of hedging and is the most common form of risk modification for financial organizations.

price of the underlying and the price of the derivative are so closely related, derivatives can provide essentially the same exposure as the underlying but can do so at lower cost and capital requirements. As such, derivatives permit the efficient shifting of risk across the probability distribution and from one party to another. One can hold the underlying and take an offsetting position in the derivative or vice versa. Whereas insurance can be designed to perform similarly, insurance functions primarily through the pooling of diverse risks. With derivatives, risks are shifted across probability distributions or payoffs and across parties, to leave specific outcomes of the conditional probability distribution with the parties most willing to bear the risk.

There are several types of derivatives, and the manner in which they provide risk shifting varies by type. Derivatives are classified into two categories: forward commitments or contingent claims. Forward commitments are agreements that obligate two parties to undertake a transaction at a future date at a price or rate agreed on when the commitment is made. Forward commitments include such instruments as forward contracts, futures contracts, and swaps. Forward commitments can be used to lock in a future interest rate for a borrower or lender, the purchase or sale price of an asset, or an exchange rate for a currency. Parties who engage in forward commitments do not pay any money at the initiation of the contract. In lieu of any up-front payment from one party to the other, the two parties agree on the terms of the transaction that will be consummated at the end of the contract. Depending on movements in the price or rate of the underlying, one party will ultimately gain from the transaction while the other will lose or, in the less likely case, both parties could breakeven. For example, a corporate treasurer can use a forward contract to lock in the rate at which a foreign cash flow will be converted into the company's domestic currency. Regardless of movements in the exchange rate during the life of the contract, the foreign cash flow will convert to the domestic currency at a rate that is locked in when the contract is initiated. On the opposite side of the transaction, the party can be a speculator who simply bears the risk, or it can be a dealer who intermediates the risk between the hedger and the speculator. We will discuss dealers in more detail in a few paragraphs.

EXAMPLE 4

Risk Shifting: Foreign Exchange Risk and Forward Commitments

You are a UK investor, investing 60% in a UK index fund tracking the FTSE 100, the leading index of the UK equity market, and 40% in US Treasuries. You expect returns in US dollars on the US Treasuries of 1.6%, and because you expect 0% return on the USD/GBP exchange rate, this return expectation equals your return expectation in sterling. You expect a return of 5.5% on the FTSE 100. Hence, the expected return on your portfolio is 3.9%.

$$R_P = w_1 \times R_1 + (1 - w_1) \times [(1 + R_{Ic}) \times (1 + R_{FX}) - 1].$$

$$R_P = 0.6 \times 0.055 + 0.4 \times [(1 + .016) \times (1 + 0.0) - 1] = 0.039 = 3.9\%.$$

Risk (standard deviation of returns) for the FTSE 100 is 13.2%, the risk on US Treasuries in sterling is 11.0%, and the risk of USD/GBP is 9%. Correlations between US Treasuries, the FTSE 100, and the USD/GBP exchange rate are shown in the following correlation matrix.

	FTSE 100	US Treasuries	USD/GBP
FTSE 100	1.00	–0.32	–0.06
US Treasuries	–0.32	1.00	0.33
USD/GBP	–0.06	0.33	1.00

Using the information provided above, we can calculate the risk of the portfolio with UK large-capitalization equities and US Treasuries as follows:

$0.090)^{1/2}$

$\sigma_p = 0.0894 \approx 8.9\%$.

The risk of this portfolio is 8.9%.

As an investor, you would like to examine possibilities to reduce the risk to this portfolio without giving up too much return, and you suspect that the volatility of the USD/GBP exchange rate contributes to risk while not adding to return. You want to know the impact of hedging the currency exposure of the US Treasury investment by means of entering into a forward contract selling USD into GBP in one year's time. You want to hedge 100% of the currency risk of the US Treasury investment.

You have been told that the forward price of a currency expressed in another currency is equal to the current spot price, corrected for the difference between the deposit rates of the countries involved over the time period ("tenor") of the forward contract. The current spot price of USD in sterling is GBP0.7040. The one-year forward price is 0.7038, as 12-month deposit rates in the United Kingdom and the United States are very close to one another.

1. What would be the impact on the risk of the portfolio if you hedge 100% of your US Treasury investment's value using this one-year forward contract?

   Hide Solution

   Solution

   In a perfect hedge, the impact is the same as ignoring the currency exposure in the US Treasury investment altogether. Filling out the standard deviations and correlations as above in the following formula gives

   $$\sigma_p = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2}\ .$$

   $$\sigma_p = \sqrt{(0.6^2 \times 0.132^2) + (0.4^2 \times 0.040^2) + 2 \times 0.6 \times 0.4 \times (-0.32) \times 0.132 \times 0.040}$$

   $\sigma_p = 0.0757 \approx 7.6\%$.

   This compares to 8.9% risk in the unhedged portfolio, a substantial reduction of risk.

2. Does the decision to 100% hedge the foreign exchange exposure have any consequences for the expected return on the hedged portfolio relative to the unhedged portfolio?

   Hide Solution

   Solution

   The forward price of the currency will play a role in the return expected on the hedged portfolio. The forward price of USD in GBP is 0.7038, which is 0.03% lower than the spot price today. Because you hedge 40% of current portfolio value, the impact on your expected return is 40% × –0.03%, reducing the hedged portfolio return to 3.87% = 3.9% – 0.03%.

3. Will the proposed transaction eliminate all currency risk associated with the US Treasury investment?

   Hide Solution

   Solution

   No, unfortunately not. As time progresses, the US Treasury investment will fluctuate in value and by the end of the tenor of the forward contract will be worth less or more in USD than the amount of USD you have sold. You have

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you are under- or overhedged is subject to currency risk. The risk of being under- or overhedged can be managed by comparing, at regular intervals, the notional value of the forward position with the value of the investment and correct the hedge as required. The decision on how often to evaluate this is a trade-off between costs (monitoring costs, transaction costs) and risk.

The other type of derivative is a contingent claim, which is commonly known as an option. An option is a contract between two parties that gives one party the right to buy or sell an underlying asset or to pay or receive a known underlying rate. An option takes the form of either a call option, which provides the right to buy the underlying or to pay a known rate, or a put option, which provides the right to sell the underlying or to receive a known rate.

With a forward commitment, both parties are mutually obligated to each other. Because an option grants the right, but not the obligation, to one party, that party has an advantage over the other. Consequently, that party, the buyer of the option, must pay cash, called the premium, to the seller of the option at the start of the contract. Once the premium is paid, the option buyer has no further obligation. He can either exercise the option or he can let the option expire unexercised. In the latter case, the option buyer incurs a loss equal to the premium. If the option is a call and it is exercised, the buyer pays the fixed price or rate and receives the underlying. If the option is a put and it is exercised, the buyer receives the fixed price or rate and delivers the underlying.[28] If the buyer of the option does exercise it, he may achieve a gain that exceeds the premium paid but the gain could also be less than the premium paid, thereby resulting in a net overall loss. An option buyer could be using the option to speculate on an upward move in the underlying if a call or downward move if a put. Alternatively, the option buyer could be hedging. In the example used earlier for forward commitments, the corporate treasurer anticipating an inflow of cash in a foreign currency could buy a put option to sell that currency, thereby converting it into his domestic cash flow at a known fixed rate. The option gives the treasurer the flexibility to not exercise it if the underlying currency rises in value. This flexibility comes at the cost of having to pay a premium at the start of the transaction, thus shifting the financial outcome across the entire probability distribution of that uncertain currency rate. In contrast, with the forward contract, the treasurer does not have to pay cash at the start but is obligated to convert at the agreed-upon rate.

Derivatives can be created in public forums, such as on derivatives exchanges, or privately between two parties. On derivatives exchanges, there are a large number of individual and institutional traders that make markets in derivatives. For private derivatives transactions, there is an extensive market of large bank and non-bank dealers willing to buy and sell derivatives. In both types of markets, these dealers assume the risk of being transferred from parties who originate the transactions. These dealers almost always restructure and transfer some portion, if not all, of the risk by finding other parties that are willing to take on that risk. Ultimately, the risk is assumed by some party willing to accept the risk, producing an economically efficient outcome for all parties.

How to Choose Which Method for Modifying Risk

Choosing which risk mitigation method to use—risk prevention and avoidance, self-insuring, risk transfer, or risk shifting—is a critical part of the risk management process. Fortunately, the methods are not mutually exclusive, and many organizations use all methods to some extent. No single method provides a clear-cut advantage over the others. As with all decisions, the trade-off is one of costs versus benefits that are weighed in light of the risk tolerance of the organization or individual.

For example, many companies that have extensive foreign operations and are, therefore, highly exposed to exchange rate risk, hedge that risk using derivatives. Some companies prefer forwards, some prefer swaps, some prefer options, and some use multiple instruments. Some companies attempt to hedge currency risk by setting up operations in

do not. Some airlines that do hedge this risk do so to a far greater degree than others. Additionally, some prefer the certainty of forwards and swaps, whereas others prefer the flexibility of options, even with the up-front cost that options require. Most insurance companies rely on their actuarial knowledge but supplement it with proactive measures, such as selling risk to other parties.

To the extent possible, most organizations should avoid risks that provide few benefits and potentially extreme costs. Reasonable, low-cost precautions against risks with few benefits should always be taken. Thus, risk prevention and risk avoidance are probably the first choice of measures, especially for risks that lie outside the core competencies of the organization and have little reasonable expectation of adding value. Nonetheless, avoidance may not be the best value for its cost. Moreover, avoiding risk may mean avoiding opportunity. Thus, an organization often cannot simply decide not to take a risk, at least not for all undesirable risks.

Organizations that have large amounts of free cash flow may choose to self-insure some risks, but few organizations have so much cash that they can afford to self-insure all risks. Some risks can potentially imperil the entire capital base. Most companies would, however, prefer to self-insure to the extent possible because self-insurance reduces the costs associated with external monitoring and gives the organization the greatest flexibility. Self-insurance and avoidance should generally be clearly addressed at the governance level and be consistent with stated risk tolerance.

Risk transfer, or the use of insurance, is a widely used risk management tactic, but it may not be suitable for many types of risks. Some risks simply are not insurable, at least not in a cost-effective way. Insurance works best when risks can be pooled, and that is not the case for many types of risks, particularly those that can affect a large number of parties at the same time. The use of risk shifting tools, such as derivatives, may not be available for all types of risks, thus limiting their use in risk mitigation. For financial risks that exceed risk appetite, risk shifting is a very common choice.

The various risk management methods are not equal in terms of the risk reduction and the risk profile that remains. For example, contingent claims, such as insurance, provide the flexibility in the form of offering opportunity to profit in one direction and have a loss reduced in the other, but they require payment of cash up front. In contrast, forward commitments lock in an outcome. In other words, they provide little flexibility, but they require no cash payment up front. The risk profile that exists when a contingent claim hedge is put in place differs significantly from the risk profile that exists when a forward commitment hedge is placed. This process requires significant understanding and discussion at all levels of the organization.

To recap, risk takers should identify risks that offer few rewards in light of potential costs and avoid those risks when possible. They should self-insure where it makes sense and diversify to the extent possible. They should consider insurance when risks can be pooled effectively if the cost of the insurance is less than the expected benefit. If derivatives are used, they must consider the trade-off of locking in outcomes with forward commitments versus the flexibility relative to cash cost of contingent claims, which can tailor the desired outcomes or payoffs by shifting the risk. Ultimately, the decision is always one of balancing costs against benefits while producing a risk profile that is consistent with the risk management objectives of the organization.

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EXAMPLE 5

Measuring and Modifying Risk

1. From the perspective of an organization, which of the following *best* describes risk drivers?

   A. The probabilities of adverse events

Solution

C is correct. Risks (and risk drivers) arise from fundamental factors in macroeconomies and industries.

2. Which of the following concepts directly measures the risk of derivatives?

A. Probability
B. Delta and gamma
C. Beta and standard deviation

Hide Solution

Solution

B is correct. Delta and gamma are measures of the movement in an option price, given a movement in the underlying. The other answers can reflect some elements of derivatives risk, but they are not direct measures of the risk.

3. The *best* definition of value at risk is:

A. the expected loss if a counterparty defaults.
B. the maximum loss an organization would expect to incur over a holding period.
C. the minimum loss expected over a holding period a certain percentage of the time.

Hide Solution

Solution

C is correct. VaR measures a minimum loss expected over a holding period a certain percentage of the time. It is not an expected loss, nor does it reflect the maximum possible loss, which is the entire equity of the organization.

4. Which of the following are methods commonly used to supplement VaR to measure the risk of extreme events?

A. Standard deviation
B. Loss given default
C. Scenario analysis and stress testing

Hide Solution

Solution

C is correct. Scenario analysis and stress testing both examine the performance of a portfolio subject to extreme events. The other two answers are metrics used in portfolio analysis but are not typically associated with extreme events.

5. Which of the following is a true statement about insurable risks?

A. Insurable risks are less costly.
B. Insurable risks have smaller loss limits.
C. Insurable risks are typically diversifiable by the insurer.

Hide Solution

Solution

C is correct. Insurance works by pooling risks. It is not necessarily less costly than derivatives nor does it have lower loss limits.

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AK Example 4, Q1, how did the SD of the US treasuries in sterling change from 11% to 4%, the Sigma2. in Q2 again the effect on hedged portfoio return should be 0.4% times -0.03% and not just 3.9 - 0.03%?

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SC Hi all, in solution 2 of example 4, we didn't use the 40%*(-0.03%); why? 0.03% is deducted directly from the return in solution 1 (3.9%).

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